CAB432 CLOUD COMPUTING · ASSESSMENT 3 · CLOUD PROJECT

Battle Snake: a horizontally scaled, server-authoritative multiplayer Slither.io on AWS

Technical report

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AWS region / account	ap-southeast-2 · 901444280953
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Generative AI use declaration [STUDENT TO REVIEW AND CONFIRM BEFORE SUBMISSION]

Generative AI was used extensively in this assessment. Anthropic's Claude (through the Claude Code agentic coding tool) worked under my direction and on my instructions: it inspected the starting repository, designed the architecture, wrote the server, client, Lambda, infrastructure-as-code and test code, ran the local, Docker and AWS deployments and tests, collected the evidence in the `evidence/` folder, and drafted this report from that evidence. I set the requirements, scope and constraints, provided the account access and the starting code (a fork of Loonride's open-source Slither.io tutorial clone), reviewed the outputs, and am responsible for the submitted work.

All measurements reported here come from the deployed system and test runs, not from the AI's assumptions. ⟦Replace or extend this statement with the exact wording QUT/CAB432 requires, and describe your own review and contribution accurately.⟧

Executive summary

This project turns Loonride's single-player Phaser Slither.io tutorial into a cloud-native multiplayer game, deployed on AWS in ap-southeast-2 as a vehicle for demonstrating scalable cloud architecture. The original sprites, renderer and gameplay rules are reused, but every outcome (movement, eating, growth, collision, death, score) is now decided by a server-side simulation. Browsers send only steering intent over WebSockets and render interpolated snapshots.

The unit of scale is an **independent authoritative world**: one ECS Fargate task runs one 50-snake world for up to four humans, with NPCs filling the remaining slots. Worlds share no real-time state, so capacity grows linearly by adding tasks (`worlds = ceil(humans / 4)`), while everything of lasting value leaves the task as events (SQS → Lambda → DynamoDB). CloudFront is the single HTTPS/WSS entry point: it serves the fingerprinted client from a private S3 bucket, caches the leaderboard for 10 s, and forwards WebSockets to an Application Load Balancer that accepts only CloudFront's secret header.

Key measured results:

- **ECS Service Auto Scaling** took the service from **1 to 5 tasks** in seven minutes (12:10–12:17 AEST) under 14 synthetic players, seating all of them across four worlds plus one empty headroom world, with 0 connection errors and RTT p50 16 ms. Measurement showed a full world uses only 20–30 % of its 0.25 vCPU, so a custom world-occupancy metric (target 75 %) is the primary scaling signal, with CPU as a safety net. ECS task scale-in protection prevents scale-in from ending games in progress. When the load stopped, it scaled back in step by step to the minimum of one task (12:47–12:59).
- **Edge caching:** fingerprinted assets are CloudFront hits from the Brisbane edge (≈20 ms). The leaderboard API drops from 615 ms (miss) to ≈20 ms (hit).
- **Asynchronous, idempotent persistence:** real game events flowed through SQS and Lambda into DynamoDB, including out-of-order delivery. A poison message was retried three times, dead-lettered, and raised an alarm.
- **Resilience:** stopping a task that hosted four players moved them to other worlds within one 5 s sample. ECS replaced the task in under a minute, and the dying task flushed all 206 of its events.
- **Cost:** measurement identified egress, not compute, as the main cost. Delta encoding, half-resolution bodies and permessage-deflate cut measured egress per player by ~80 % (≈ 105 → 21 KiB/s, deployed), giving ≈ US$0.014 per player-hour (≈ $630/month at 60 concurrent players).

The CAB432 account's guardrails (no IAM role, security group or certificate creation; no deletion of buckets or log groups) forced several adaptations, which are documented with their residual risks in §3.2.

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1 Introduction

Real-time multiplayer games are a demanding workload for cloud architecture. They need stateful compute that updates many entities dozens of times per second, low-latency bidirectional communication, durable records of outcomes, and capacity that follows a player population that changes through the day. This project uses a small Slither.io-style game as a vehicle to build and measure such a system on AWS.

The starting point was Loonride's open-source tutorial clone of Slither.io (Loonride, 2017). It is a single-browser Phaser CE 2.6.2 game in which a P2 physics engine running in the page decides movement, collisions, food consumption and growth for one player and two bots. Nothing in it is networked, and the browser is the authority on every outcome, so a multiplayer version could not trust it. The project keeps the original sprites, renderer, movement constants and rules, and moves all authority to a server-side simulation.

Scope

- Players enter an anonymous display name and join a **world**: an arena that always holds **50 snakes**. At most **4 are humans** and the rest are server-controlled NPCs (`NPCs = 50 − humans`).
- The first human starts a lobby of **at most 10 seconds**. The world starts early when four humans are present. Later humans replace an NPC immediately. A fifth human receives `ROOM_FULL` and is routed elsewhere.
- The server simulates movement, eating, growth, collisions and death at 30 Hz. Clients only send intent (heading and boost) and render interpolated snapshots.
- Deaths and session results are stored durably and feed a global leaderboard.

Objectives

The engineering objectives were a cloud architecture that (i) scales **horizontally with demand using non-Lambda auto-scaling**, (ii) distributes connections across stateful workers, (iii) keeps durable data outside those workers, (iv) caches static content at the edge, (v) processes outcomes asynchronously and idempotently, (vi) is observable and recovers from failures, and (vii) stays cost-proportionate. Every claim in this report is backed by a captured artefact. Evidence identifiers such as **[E11]** refer to the register in Appendix B and files in the repository's `evidence/` folder.

2 Vendor-agnostic architecture

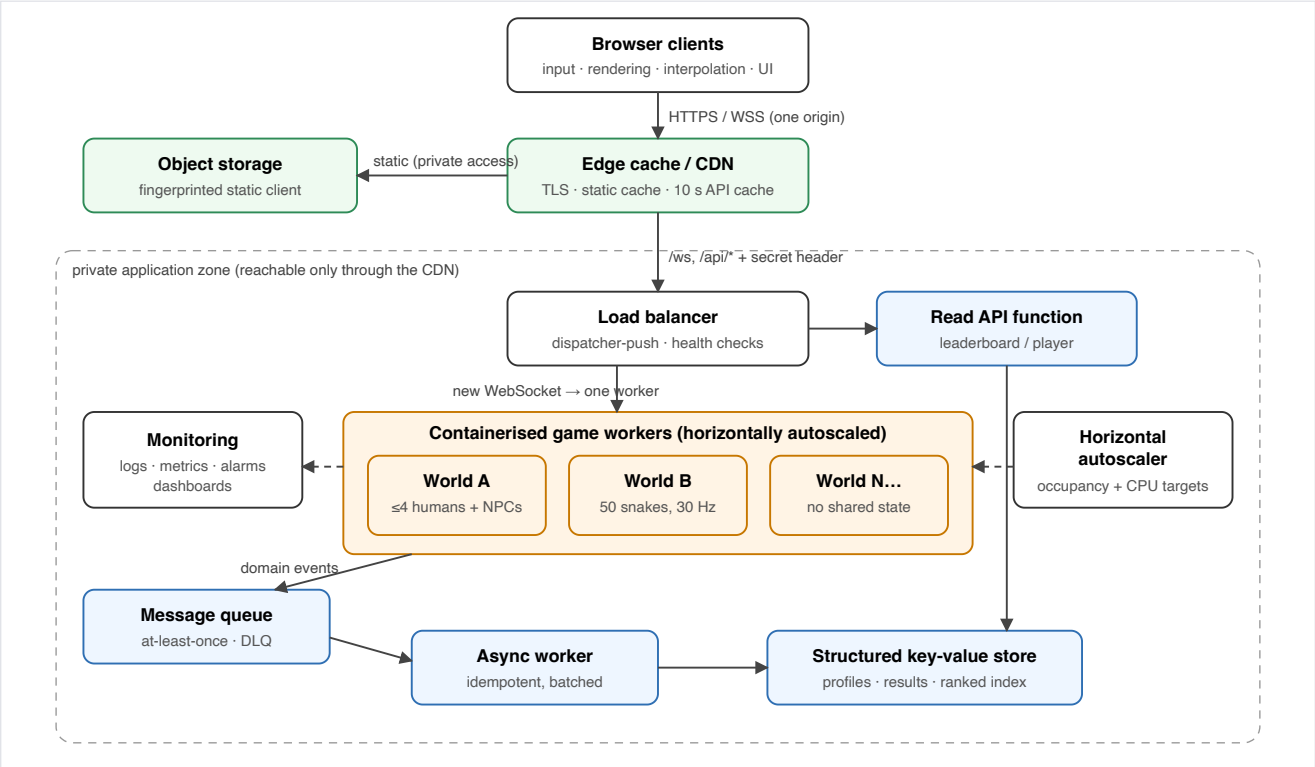


Figure 1. Vendor-neutral architecture. Orange = stateful game workers; blue = asynchronous/serverless path; green = edge delivery. Only the CDN is publicly reachable.

2.1 Components

Concept	Responsibility in this system	Why it is needed
Edge cache / CDN	Terminates TLS near players. Caches fingerprinted static files for a year and the leaderboard response for 10 s. Proxies WebSocket upgrades and API calls to the private origin.	Static delivery scales independently of game servers. Repeat requests never reach the origin. A single origin for page, API and socket removes CORS and needs one certificate.
Object storage	Holds the built client (HTML, one JS bundle, Phaser, sprites), readable only by the CDN.	Durable, cheap, serverless hosting for immutable assets.
Load balancer	Accepts traffic only from the CDN, health-checks workers, and assigns each new WebSocket to a healthy worker (dispatcher-push).	Hides individual workers, removes failed ones from rotation, and spreads *new* sessions.
Containerised game workers	Each worker process owns exactly one authoritative world: lobby, 50 snake slots, food, 30 Hz tick, 15 Hz snapshots.	Real-time simulation needs long-lived in-memory state, held sockets and a steady CPU budget. Short-lived functions fit this poorly.
Horizontal autoscaler	Adds or removes workers from measured world occupancy and CPU. Workers hosting players opt out of scale-in.	Capacity follows demand without terminating games in progress.
Message queue + dead-letter queue	Durable buffer for domain events (`PLAYER_DIED`, `SESSION_ENDED`, `WORLD_STARTED`, `WORLD_COMPLETED`).	Decouples the latency-critical tick from database writes, absorbs bursts, retries failures and isolates poison messages.
Asynchronous worker (function)	Applies events in batches, idempotently.	Scales to zero. Retry semantics come from the queue.
Structured key-value store	Player profiles, death history, world records, idempotency markers, and a secondary index ordered by best score.	Durable state lives outside the stateful workers. Every access is a key lookup or an ordered range, never a scan.
Read API (function)	Serves the leaderboard and a player's statistics.	Keeps workers focused on simulation. Responses are cacheable at the edge.
Monitoring system	Structured logs, platform and custom metrics, dashboards, alarms.	Drives scaling decisions and makes failures diagnosable.

2.2 Independent authoritative worlds

The central decision is that **the unit of scale is an independent world**. A world is small (50 snakes, ≤4 humans), lives entirely in one worker's memory, and shares no real-time state with any other world. Scaling out means *adding worlds*, not partitioning one world across machines.

This removes distributed consistency from the hot path. Each 30 Hz tick is a single-threaded loop over local memory (a full world's median tick measured 1.3 ms on a 0.25 vCPU Fargate task, §4.5), with no cross-node locks, replicated physics or consensus protocol. It also makes capacity planning arithmetic: because NPCs are workload rather than users,

```
requiredWorlds = ceil(concurrentHumans / 4)        runningWorkers ≈ ceil(concurrentHumans / (4 × targetOccupancy))
```

The trade-off is explicit. **A world's transient state exists only inside its worker**. If that worker fails or is replaced, the positions and current lengths in that world are lost, and its players reconnect to another world within seconds. That is acceptable for a casual arena game, where a round is short and losing it costs a player one life. Everything with long-term value (scores, deaths, best scores, world summaries) leaves the worker as an event at the moment it happens and is stored durably. Restoring exact world state would need checkpointing or replication at tick rate, a large cost in complexity, bandwidth and latency for little player benefit. A single global world was rejected for the same reason: it would need spatial partitioning with entity hand-over and cross-node collision detection.

2.3 Data and control flows

1. **Join.** Browser → CDN → load balancer → some worker. If the world has a free human slot, the player joins. Otherwise the worker replies `ROOM_FULL` and closes. The client retries after 150–600 ms jitter, so the load balancer can choose another worker. After five quick attempts it backs off exponentially (≤5 s) for about three minutes while the autoscaler adds a world.
2. **Lobby and start.** The first human starts a ≤10 s countdown. The fourth human starts the world immediately. At start, empty slots are filled with NPCs.
3. **Play.** Clients send intent at ≤20 Hz. The worker ticks at 30 Hz and sends each client a 15 Hz snapshot limited to its area of interest. Clients render 110 ms behind real time and interpolate.
4. **Persist.** Outcomes become events with unique IDs → queue → worker function → conditional, transactional writes → ranked index → read API → edge cache.
5. **Scale.** Workers publish occupancy and tick metrics. The autoscaler tracks targets for occupancy and CPU. Workers with players protect themselves from scale-in.

2.4 Alternatives considered

Decision	Chosen	Rejected	Reasoning
Real-time transport	WebSocket	HTTP polling; WebRTC/UDP	Polling costs one request per update per client at 15 Hz. WebRTC needs signalling and TURN servers and isn't proxied by standard CDNs or load balancers. WebSocket gives one full-duplex connection that passes through the CDN and load balancer.
Authority	Server-authoritative	Client-authoritative (original game)	A client could forge position, score or invulnerability. The leaderboard is only meaningful if the server is the referee.
World topology	Many small independent worlds	One global world	Linear scale-out without distributed state (§2.2).
Game compute	Managed containers	Functions; self-managed VMs	Functions are short-lived and stateless. VMs add patching and capacity management, and only pay off at sustained high utilisation.
Persistence path	Async queue	Synchronous writes from the game loop	Synchronous writes add network latency and database throttling to the tick and connection path. A queue absorbs bursts and retries.
Static delivery	CDN + private object store	Served by game workers; public bucket	Workers would spend CPU and bandwidth on static files. A public bucket has no edge cache and more exposure.
Datastore	Key-value store + ordered index	Relational database	Access patterns are key lookups and "top N by score". Pay-per-request with no instance to run. Joins are unused.
Scaling signal	World occupancy (custom) + CPU	CPU only; request count	Request count is meaningless for long-lived sockets. Measurement showed a full world uses only ~20–30 % of its CPU allocation, so CPU alone never triggers (§4.5). Occupancy measures user capacity directly.
Lost world	Reconnect to any world	Checkpoint/replicate state	Disproportionate cost for casual sessions. Durable stats are already safe.

3 Vendor-specific implementation (AWS)

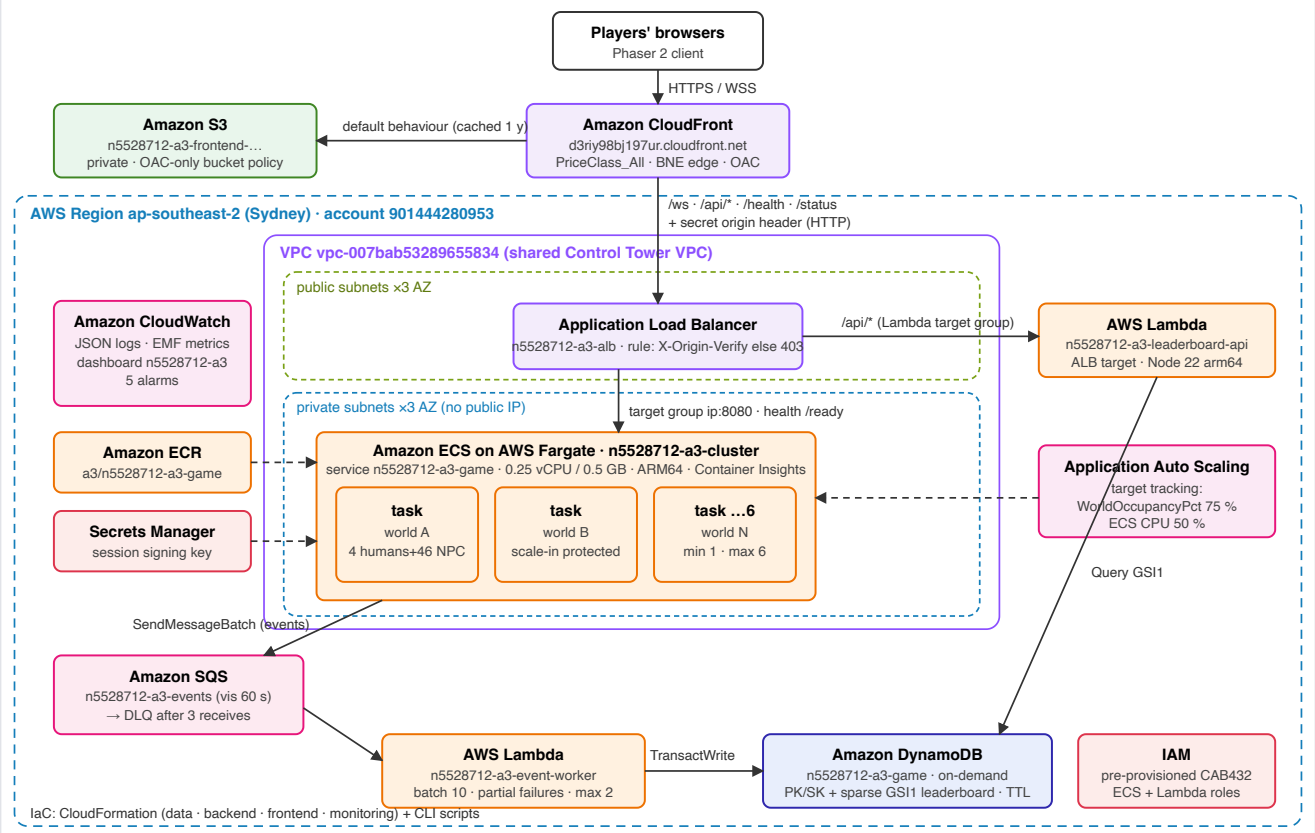


Figure 2. Deployed AWS architecture (ap-southeast-2). Resource names are the real deployed identifiers. The ALB sits in the three public subnets; game tasks run in the three private subnets without public IPs.

3.1 Service mapping

Concept	AWS service	Deployed configuration and rationale
Edge cache / CDN	Amazon CloudFront	One distribution (`d3riy98bj197ur.cloudfront.net`), HTTP/2 and HTTP/3, `PriceClass_All` (Australian edge locations are only in this class; requests were served from `BNE50`, Brisbane). Default behaviour → S3 with the managed *CachingOptimized* policy. `/ws` → ALB with *CachingDisabled* and *AllViewerExceptHostHeader*, which forwards the WebSocket upgrade headers. `/api/*` → ALB with a custom policy (default TTL 10 s, cache key = `limit`, `id` query strings only). The origin request carries a secret `X-Origin-Verify` header.
Object storage	Amazon S3	Private bucket `n5528712-a3-frontend-901444280953`, Block Public Access on, bucket policy allowing `s3:GetObject` only to this distribution through Origin Access Control (OAC). Objects carry `Cache-Control: public, max-age=31536000, immutable` (content-hashed names) or `no-cache` (`index.html`).
Load balancer	Application Load Balancer	`n5528712-a3-alb` across three AZs, idle timeout 120 s. Listener rules forward `/ws`, `/health`, `/ready`, `/status` to an IP target group (port 8080, health check `/ready` every 10 s, deregistration delay 15 s, round robin) and `/api/*` to a Lambda target group, **only when the secret header is present**. The default action is a fixed 403.
Game workers	Amazon ECS on AWS Fargate	Cluster `n5528712-a3-cluster` (Container Insights on), service `n5528712-a3-game`. Task: 0.25 vCPU, 0.5 GB, **ARM64 (Graviton)**, image from ECR, container health check on `/health`, `stopTimeout` 30 s, deployment circuit breaker with rollback. Tasks run in private subnets with no public IP.
Container registry	Amazon ECR	`a3/n5528712-a3-game`, scan on push, lifecycle keeps the newest 10 images. Image built for `linux/arm64`, `node:22-alpine`, non-root.
Autoscaler	Application Auto Scaling (ECS Service Auto Scaling)	Scalable target 1–6 tasks. Target tracking on the custom metric `WorldOccupancyPct` (Average, target 75 %) and on `ECSServiceAverageCPUUtilization` (target 50 %). Scale-out cooldown 60 s, scale-in 180 s. Tasks with players set **ECS task scale-in protection** through the ECS agent endpoint.
Message queue	Amazon SQS	Standard queue `n5528712-a3-events`: visibility timeout 60 s (6× the worker timeout), retention 4 days, long polling 20 s, SSE. Redrive to `n5528712-a3-events-dlq` after `maxReceiveCount = 3` (retention 14 days).
Async worker	AWS Lambda	`n5528712-a3-event-worker`, Node.js 22, arm64, 256 MB, 10 s timeout. SQS event source mapping: batch 10, 2 s batching window, `ReportBatchItemFailures`, maximum concurrency 2.
Read API	AWS Lambda (ALB target)	`n5528712-a3-leaderboard-api`, 128 MB. Returns `Cache-Control: public, max-age=10` for the leaderboard.
Structured store	Amazon DynamoDB	`n5528712-a3-game`, on-demand, SSE, TTL on `ttl`, sparse GSI `GSI1` (§3.4).
Configuration and secrets	Secrets Manager, SSM Parameter Store, environment	The HMAC key that signs anonymous player IDs is a generated Secrets Manager secret, read by tasks at start-up with the task role. The CloudFront→ALB header value is an SSM SecureString. All other settings are task-definition environment variables. The image contains no credentials or secrets.

Monitoring	Amazon CloudWatch	JSON logs. Custom metrics through the Embedded Metric Format (namespace `n5528712/A3Game`: occupancy, humans, entities, tick p50/p99, snapshot bytes, worlds started/ended, events dropped). Dashboard `n5528712-a3`. Five alarms (DLQ not empty, worker errors, queue backlog, unhealthy targets, slow tick). CloudFront additional metrics (cache hit rate).

3.2 Account constraints and how the design adapted

The shared CAB432 account applies explicit IAM deny policies to the student role. They were discovered during deployment, recorded in `docs/current-state.md`, and each forced a concrete design change:

Denied action (verified)	Consequence	Adaptation
`iam:CreateRole`	Roles can't be defined in the templates.	The pre-provisioned `Execution-Role-CAB432-ECS`, `Task-Role-CAB432-ECS` and `CAB432-Lambda-Role` are used. Their policies can't be read, so least privilege can't be demonstrated (a stated limitation). Features were verified empirically: SQS send, Secrets Manager read and task protection from tasks; SQS consume and DynamoDB from Lambda.
`ec2:CreateSecurityGroup`	No dedicated SGs restricting the ALB to CloudFront or the tasks to the ALB.	The shared `CAB432SG` is used. Compensating controls: (1) tasks in **private subnets with no public IP**, so they are unreachable from the internet; (2) the ALB forwards only requests carrying the secret header that only CloudFront adds. Direct ALB requests receive 403 **[E19]**.
`acm:RequestCertificate`	No TLS certificate for an ALB origin hostname.	Viewer ↔ CloudFront is HTTPS/WSS. CloudFront ↔ ALB is HTTP (a documented residual risk; with a certificate the origin policy would change to `https-only`).
`s3:DeleteBucket`, `s3:PutEncryptionConfiguration`, `s3:PutLifecycleConfiguration`	The first data-stack deployment could not roll back its bucket (`ROLLBACK_FAILED`).	Buckets are created idempotently by `ensure-buckets.sh` and referenced by the stacks. Default SSE-S3 encryption still applies. New buckets have S3 attribute-based access control, so tags are applied with `s3control tag-resource`.
`logs:DeleteLogGroup`, `logs:PutRetentionPolicy`	Same rollback problem; retention can't be set.	Log groups are created by script. Retention stays at the default (a cost note in §5).
`lambda:GetFunction` on not-yet-existing functions	CloudFormation's Lambda resource handler fails before creating the function.	Functions, the ALB invoke permission and the SQS event source mapping are created or updated by `deploy-lambdas.sh` (tagged `create-function` is permitted). The stack references the function ARN.

3.3 Security

- **Public vs private access.** Only CloudFront is a public entry point. S3 is OAC-only (direct URL → 403). The ALB answers 403 without the origin secret. Tasks have no public IP **[E2, E19]**.
- **Transport.** HTTPS/WSS between player and edge, with HTTP redirected to HTTPS. The internal hop is discussed in §3.2.
- **Untrusted clients.** The server accepts only four message types (`join`, `in`, `respawn`, `ping`), validates every field (finite numbers, ranges, UUID/hex formats), caps message size at 1 KB, sanitises and truncates display names to 16 characters (rendered with `textContent`), rate-limits each socket with a token bucket (40 msg/s, sustained abuse closes the socket), and ignores any position, score or length a client sends. Unit and integration tests cover forged fields and malformed JSON.
- **Identity without accounts.** Anonymous player IDs are server-generated UUIDs signed with an HMAC key held in Secrets Manager. A client presenting an ID with a wrong signature gets a fresh ID, so leaderboard entries can't be hijacked (test: "*forged id is replaced*").
- **Credentials.** No access keys exist anywhere: tasks and functions use IAM roles, and deploy scripts use the operator's SSO session.

3.4 Data model

Entity	PK	SK	Notable attributes
Player profile	`PLAYER#<id>`	`PROFILE`	displayName, bestScore, deaths, kills, sessions, playMs. `GSI1PK='LB'`, `GSI1SK=bestScore` written only once a score exists (sparse index).
Death record	`PLAYER#<id>`	`DEATH#<time>#<eventId>`	score, length, kills, killedBy, survivedMs, worldId
World record	`WORLD#<worldId>`	`META`	start/end times, humans, NPCs, lobby wait, peak humans, end reason
Idempotency marker	`EVENT#<eventId>`	`EVENT`	type, processedAt, `ttl` = now + 7 days

Access patterns: leaderboard = `Query GSI1 (GSI1PK='LB', descending, Limit N)`; player stats = `GetItem`; history = `Query PK, begins_with(SK,'DEATH#')`. No access pattern uses a Scan. The leaderboard partition key is a single value. At far more than this project's write rate (≫1,000 writes/s) it would be write-sharded (`LB#0..n`) and merged on read.

3.5 Infrastructure as code and observability

Four CloudFormation stacks (`data`, `backend`, `frontend`, `monitoring`) plus idempotent CLI scripts reproduce the deployment with one command (`infra/scripts/deploy.sh`). All parameters come from environment variables, and every taggable resource carries `purpose = assessment 3` and `qut-username = n5528712@qut.edu.au`. The data stack uses `DeletionPolicy: Retain` and is separate from compute, so the backend can be torn down without losing results. (The DynamoDB table survived an early failed stack and was *imported* into the new stack rather than deleted.)

Game tasks emit one JSON log line per event with `taskId`, `worldId`, `playerId` and `eventId`, so a single event can be followed from `event_published` in the game log to `event_processed` in the worker log. Custom metrics use the Embedded Metric Format: a JSON line every 10 s that CloudWatch Logs converts into metrics, so no metrics API permission or SDK is needed. The metric dimension is the service name only, so the *Average* across tasks is the fleet-wide value the scaling policy consumes.

4 Application implementation

4.1 From a single-player tutorial to an authoritative server

The original game delegated motion to P2 physics: the head was moved with `moveForward`, its positions were pushed into a `headPath`, each of 30 body sections was placed on that path about `17 × scale` behind the previous one, and an invisible "edge" circle locked in front of the head killed the snake when it touched another snake's body. P2 is coupled to Phaser sprites and cannot run headless. The server therefore re-implements the same model directly (`server/src/game/snake.js`): the same constants (start scale 0.6, 30 sections, 130/200 px/s, a turn rate of 2 rad/s derived from P2's `rotateLeft(40)` and its px→m factor of 20, growth ×1.01 per food), arc-length placement of sections along the head path, and an edge-point collision test. A spatial hash grid makes collision and food queries local. A full 50-snake world ticks in about 0.3 ms on a laptop core and the 37 server tests pin the behaviour down.

The NPCs keep the original bot's "trend" wander (keep turning one way, flip with probability 1/20) and add three priorities: stay inside the arena, avoid bodies and approaching heads, and seek nearby food. Instrumenting deaths during development showed that 69 % of NPC deaths were head-to-head collisions while chasing the same food. Adding an "approaching head" check raised median NPC lifetime from 15 s to 23 s, and that was judged sufficient (the brief asks not to over-engineer AI).

The client keeps the original assets and drawing code (tinted circle sections, the `shadow.js` light-up effect while boosting, and the eye pair) but now only renders server state. Because `circle.png` is solid red, tinting could only produce reds. The client desaturates it once at start-up into a white texture, which keeps the original outline shading and lets each snake take any colour.

Original (client-authoritative)	Now
P2 physics in the browser decides position, collisions, eating, growth, death	Server tick at 30 Hz decides everything. The browser sends heading + boost only.
1 player + 2 bots, 1,600 × 1,000 px world	≤4 humans + NPCs = 50 snakes, circular arena of radius 2,400 px, 700 ambient food
Food pulled into the head by a revolute constraint	Head–food overlap test. Food deltas are applied on the client at the snapshot's render time.
No identity, score or leaderboard	Signed anonymous ID, score, in-world top 10, persistent all-time top 10

4.2 Workflows

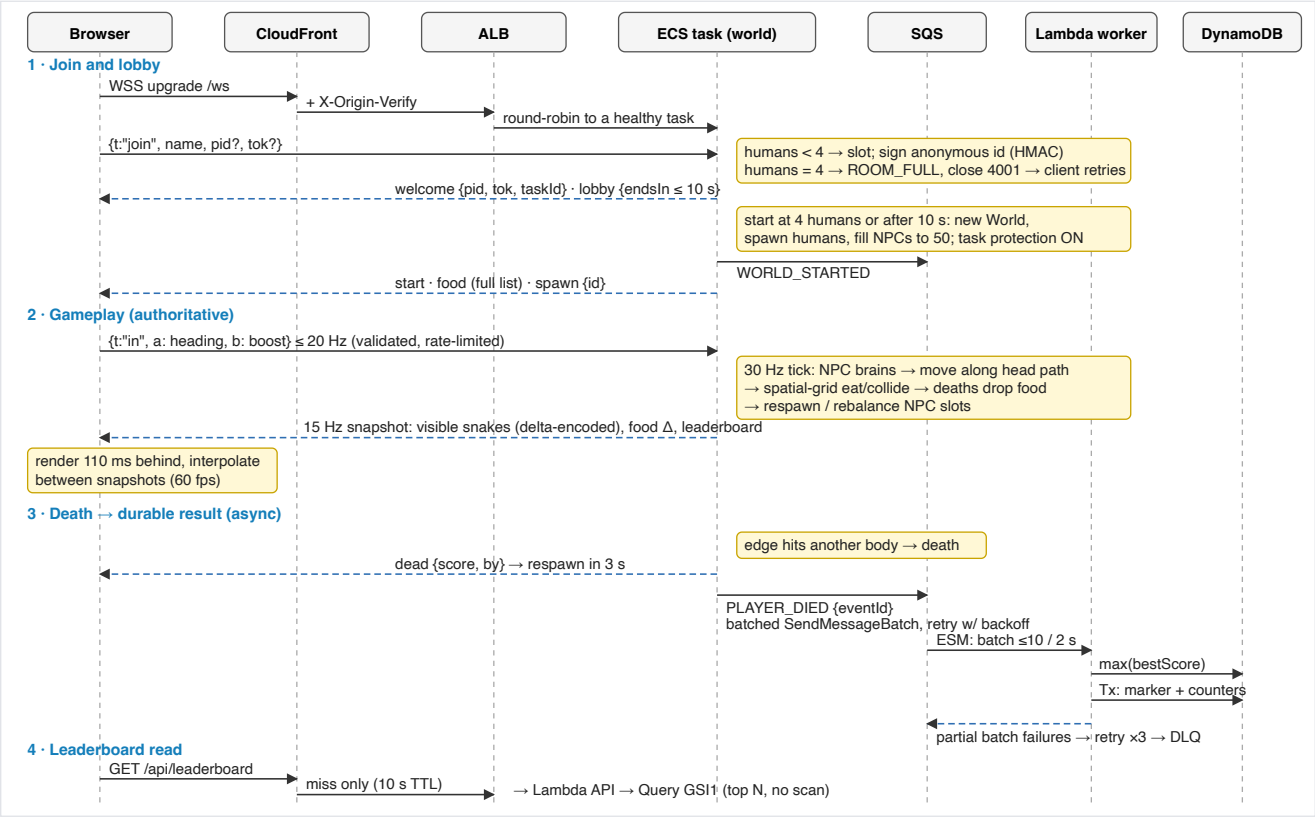


Figure 3. Join/lobby, authoritative gameplay, asynchronous persistence, and the cached leaderboard read, as implemented.

Each task's `Room` runs a small state machine: *idle* → (first human) → *lobby* → (4 humans or 10 s) → *running* → (last human leaves) → *idle*. Leaving *running* discards the world entirely, so an empty task does no simulation work (≈3 % CPU, [E10]). Every tick enforces the slot invariant `NPCs = 50 − humans`: a late joiner removes the smallest NPC, and a departing human is replaced by a new NPC. A dead human keeps their slot during the 3 s respawn timer, so the active count briefly reads 49. Unit tests check the fill for 1–4 humans (49/48/47/46 NPCs), the ≤10 s lobby, the immediate start at four humans, ROOM_FULL for the fifth, late join, and world completion [E4, E5].

4.3 Networking, interest management and bandwidth

Snapshots contain only the snakes whose bounding box intersects the player's viewport plus a 350 px margin. Each snake is serialised once per snapshot and the fragment reused for every client. Food changes travel as deltas. The client buffers snapshots, maps server time to local time using the minimum observed one-way offset, and renders 110 ms in the past, interpolating section positions between the two surrounding snapshots.

Bandwidth was measured, not assumed, and it turned out to be the dominant running cost (§5). Two optimisations followed from the measurements, both captured by the load tester:

Version (same 4-player, 40 s run, 0.25 vCPU container)	Bytes on the wire	Per player	Evidence
Absolute coordinates, JSON	14.6 MB	≈ 91 KiB/s	E14a
Delta-encoded body points (head absolute, then small offsets)	8.6 MB (–41 %)	≈ 54 KiB/s	E14b
+ WebSocket permessage-deflate (level 1, context takeover)	2.3 MB (–74 % vs. delta)	≈ 14.7 KiB/s	E17

Those runs used young worlds. In the 25-minute AWS run, snakes grew and more of them were in view, and the server-side `SnapshotBytesPerSec` metric showed **≈100–111 KiB/s per player uncompressed** [E17]. An offline model of a 4-minute-old world (`tools/bandwidth-sim.mjs`, mean snake length 75) gave 157 KiB/s raw, 47 KiB/s with deflate and **30.5 KiB/s** when bodies are also sent at **half resolution** (sections 0, 2, 4, … plus the tail; the client rebuilds the skipped ones as neighbour midpoints, within 2 px of the true positions, which a server test checks using the real client decoder). Both changes were deployed. The deployed result: 4 players for 10 minutes through CloudFront received 156.2 MiB of snapshot payload as **48.9 MiB on the wire** (–69 % from compression), **≈ 21.2 KiB/s per player**, with RTT p50 17 / p95 30 ms and 0 errors [E17]. Relative to the ≈105 KiB/s measured before these changes, that is a reduction of about 80 %.

Compression added about +2.5 points of host-core CPU (≈ +10 points of a 0.25 vCPU task), a good trade because egress costs more than compute in this design. Enabling compression also exposed a shutdown bug: sends become asynchronous, and the server terminated sockets before the final `bye` message was flushed. The integration test caught it, and shutdown now waits up to 1.5 s for close handshakes.

4.4 Asynchronous, idempotent persistence

The game loop never waits for the network to persist results. `publish()` appends an event envelope (`eventId` UUID, type, version, time, worldId, taskId, payload) to an in-memory buffer. A flusher sends batches of up to 10 with `SendMessageBatch`. Rejected entries are retried with exponential backoff and jitter, up to 4 times, and the buffer is bounded at 5,000 events (oldest dropped). An event that still can't be sent is written in full to the log (`event_dropped`), so it is recoverable. On SIGTERM the buffer is flushed within a deadline. In the resilience test the terminating task flushed 206 events with 0 dropped [E16].

SQS standard queues deliver at least once and do not preserve order, so the worker is written for both:

- **Idempotency.** Non-idempotent changes (incrementing deaths and kills, inserting a death record) run in one `TransactWriteItems` together with a conditional put of `EVENT#<eventId>`. A redelivery fails the condition and changes nothing. The best-score update is a conditional "max" (`attribute_not_exists(bestScore) OR bestScore < :s`), which is idempotent by nature, so it runs *before* the transaction: a crash between the two can never lose a high score.
- **Ordering.** World records use `SET` updates, so `WORLD_STARTED` and `WORLD_COMPLETED` commute. In the deployment a `PLAYER_DIED` was processed before its world's `WORLD_STARTED` [E8], and the result was still correct.
- **Poison messages.** Schema validation failures and transient errors are both reported as `batchItemFailures`, so only those records return to the queue. After three receives SQS moves them to the DLQ, which raises an alarm. A deliberately malformed message was rejected three times, 60 s apart (the visibility timeout), then moved to the DLQ, and the `dlq-not-empty` alarm went to ALARM [E16].

The leaderboard is read by a separate Lambda behind the ALB. It queries the sparse GSI in descending order and returns `Cache-Control: public, max-age=10`. CloudFront therefore answers repeated reads at the edge (§4.7), and at most one request per 10 s per edge location reaches DynamoDB, regardless of player count.

4.5 Scaling: design and measured behaviour

Choosing the signal from measurements

The brief suggested starting with CPU target tracking. Local measurement in a container capped at 0.25 vCPU, followed by Fargate measurement, showed why CPU alone can't drive this system. A world can never hold more than four humans, so a full world's CPU is bounded: **≈20–30 % of a 0.25 vCPU Graviton task**, while an idle task sits at ≈3 % [E10, E15]. With any sensible CPU target (40–50 %), the service would stay at one task while every world is full and new players wait. The scaling question is really "are there free seats?", so the primary policy tracks a custom metric, `WorldOccupancyPct = humans / 4 × 100`, averaged over all tasks, with a target of 75 %. Target tracking then keeps `tasks ≈ ceil(humans / 3)`, about one empty world in four. CPU target tracking at 50 % stays as a safety net (for example if world size or NPC count were raised). Application Auto Scaling scales out when either policy asks and scales in only when both agree.

Scale-in must not end games in progress. Each task enables ECS **task scale-in protection** when its world starts (renewed every 5 minutes, expiring after 15 if the process dies) and releases it when the world empties. During the test, the four tasks hosting players were protected and the newly started empty task was not [E12]. The autoscaler can therefore only remove empty worlds.

Scale-out experiment

Fourteen synthetic players (`load-tests/websocket-load.js`) joined through CloudFront (`wss://…/ws`), ramping over 30 s, for 25 minutes. They behave like real clients: they join, retry `ROOM_FULL`, steer with wandering input at 10 Hz, boost occasionally, respawn after dying, and ping every 2 s. The load is genuine gameplay. Nothing is a busy loop. Because one world takes four humans, ten players are initially rejected, which is exactly the situation the occupancy policy exists for.

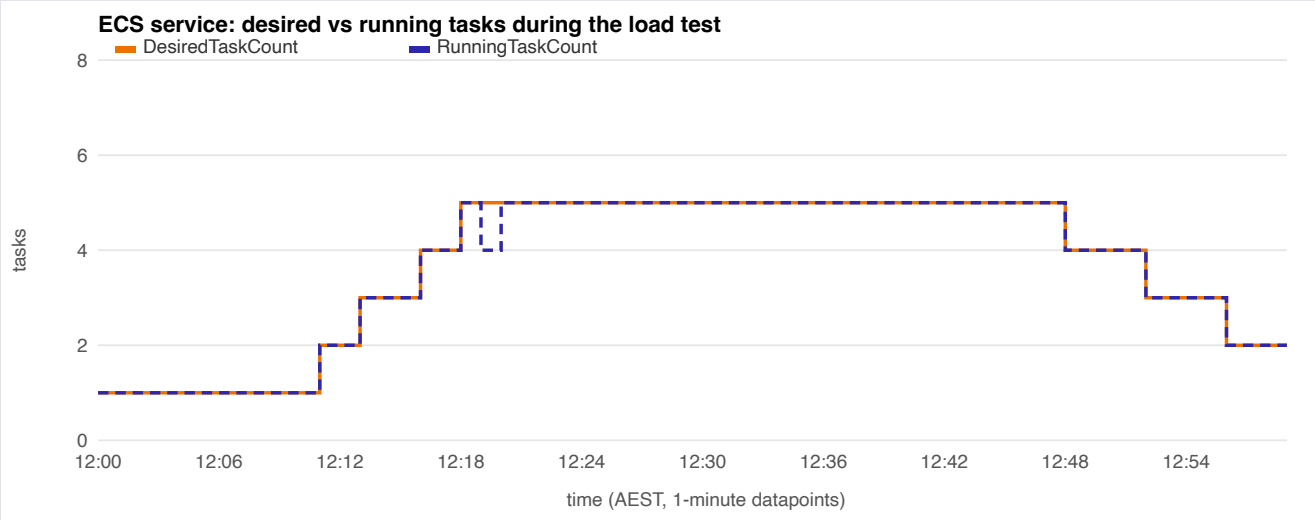


Figure 4. Desired and running tasks (Container Insights, 1-minute maximum). The dip at 12:19 is the deliberate task-failure test (§4.6).

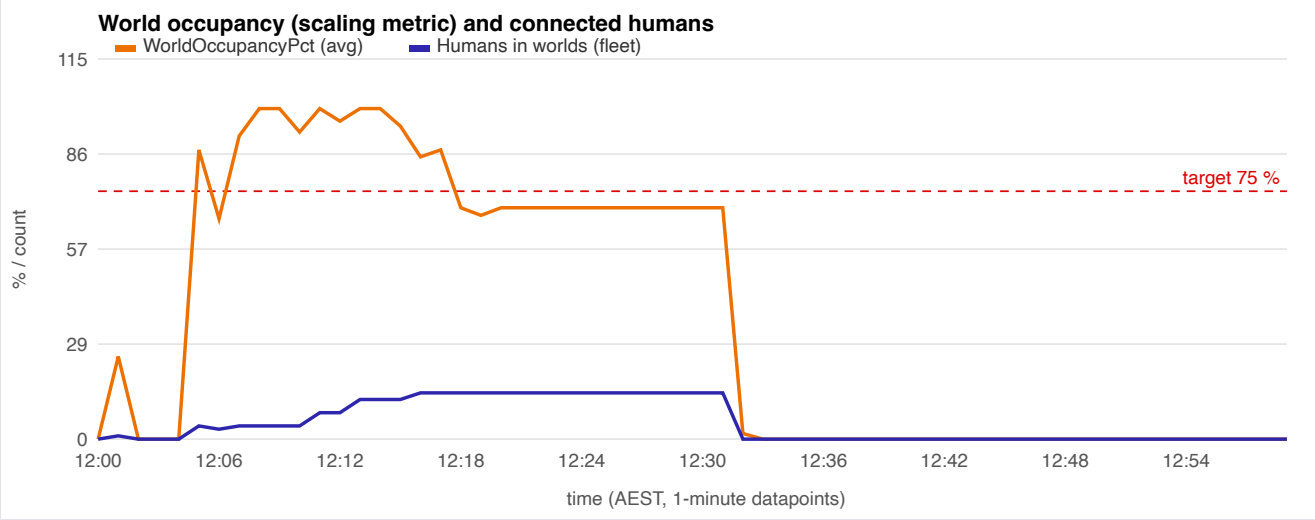


Figure 5. Fleet-average world occupancy (the scaling metric) and humans placed in worlds. An aborted first run at 12:05–12:06 appears as the early spike; the measured run starts at 12:07.

Time (AEST)	Scaling activity (Application Auto Scaling)	Players in worlds / tasks
12:07	Load starts; 1 task, occupancy → 100 %	4 in world, 10 receiving ROOM_FULL
12:10:15	"Setting desired count to 2." Task running and healthy ≈ 40 s later	8 / 2
12:12:15	"Setting desired count to 3."	12 / 3 (4 + 4 + 4)
12:15:15	"Setting desired count to 4."	14 / 4 (4 + 4 + 4 + 2): every player placed
12:17:15	"Setting desired count to 5." Occupancy 14/20 = 70 %, inside the target band, so steady	14 / 5 (one empty world as headroom)
12:32	Load stops; worlds empty; every task releases scale-in protection **[E11b]**	0 / 5
12:47:07	After 15 one-minute datapoints below 67.5 % (the policy's scale-in threshold, 0.9 × target): "Setting desired count to 4."	0 / 4
12:51:20 · 12:55:20 · 12:59:20	"… to 3", "… to 2", "… to 1": one step per evaluation after the 180 s scale-in cooldown	0 / 1 (minimum)

Scale-out is deliberately faster than scale-in: three one-minute datapoints above target versus fifteen below, and a 60 s versus 180 s cooldown. A short dip in players therefore does not destroy worlds. Each scale-out step needed three consecutive one-minute datapoints above target plus the 60 s scale-out cooldown, so the fleet grew by one world roughly every 2–3 minutes, and the sequence 1→2→3→4→5 followed `ceil(tasks × occupancy / 75)`. New tasks were registered and healthy 30–45 s after each decision. Across the run the load tester recorded **0 connection errors**, WebSocket RTT **p50 16 ms / p95 29 ms / p99 68 ms** (8,769 pings) from Brisbane via the BNE edge, and a snapshot interval of p50 67 ms (15 Hz) **[E14]**. Per-task CPU stayed at 20–30 %. On Fargate the median tick took 1.3 ms and the p99 averaged 2.3 ms (worst 10 s window 13.4 ms) against the 33 ms budget **[E15]**.

Load distribution. The ALB uses round robin for new connections. Combined with `ROOM_FULL` and client retry, this acts as simple decentralised matchmaking: the load tester's per-task counts show players spread evenly over the new worlds as they appeared (4/4/4/2) **[E14]**, with targets healthy in all three AZs **[E13]**.

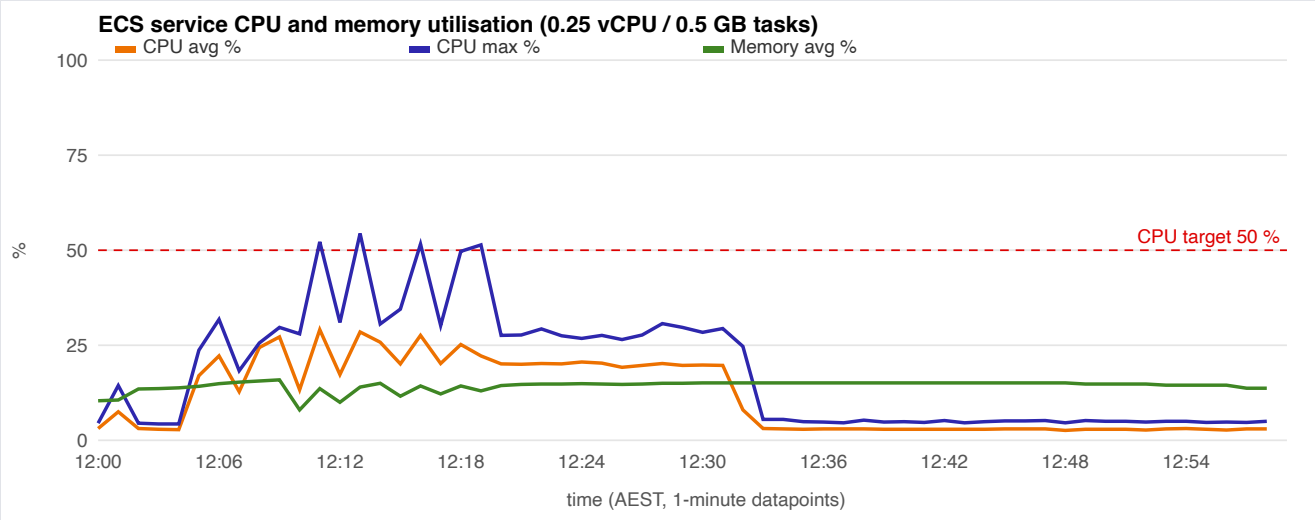


Figure 6. ECS service CPU and memory utilisation: a full world uses ≈20–30 % of its 0.25 vCPU allocation and ≈10 % of 512 MB.

4.6 Resilience

Failure	What happened (measured)
Game task lost during play (task stopped while hosting 4 players)	12:18:17 stop issued. 12:18:59 target deregistered and a replacement task started. ≈12:19:15 the ALB closed the four WebSockets at the end of the 15 s deregistration delay. All four players rejoined other worlds within one 5 s sample (two filled a world with free seats, two went to the idle headroom world). The old task completed its world, published four `SESSION_ENDED` events, received SIGTERM at 12:19:24, flushed **206 events with 0 dropped** and exited with code 0. The replacement was healthy at 12:19:28 and the service reached a steady state at 12:19:47 **[E16]**.
Poison message	Received three times, 60 s apart, logged as `event_malformed` each time, then moved to the DLQ. The DLQ alarm fired **[E16]**.
Duplicate or out-of-order delivery	Idempotency marker and commutative updates (unit-tested; out-of-order delivery observed in production) **[E8]**.
Transient AWS errors	Publisher: per-entry retry with exponential backoff and jitter, bounded at 4 attempts and a 5,000-event buffer. Worker: the failed record returns to the queue (unit-tested).
Hung tick loop	`/health` fails if no tick for 2 s. The container health check and ECS replace the task.
Full worlds	Client: five fast jittered retries, then bounded exponential backoff (≤5 s) for ~3 minutes, never forever. The load test verified that waiting players were seated as capacity arrived.

One finding changed the understanding of graceful shutdown. Because ECS drains the ALB target *before* sending SIGTERM, the ALB, not the application, ends WebSocket sessions when the deregistration delay expires. The server's `bye` message is only delivered when the process itself initiates shutdown. Resilience therefore rests on the client's reconnect path, and the measurement shows that path is fast.

4.7 Edge caching

`tests/cache-test.sh` requested the same objects five times from Brisbane **[E6]**:

Object	Origin	Result
`app.<hash>.js` (immutable, 1 year)	S3 via OAC	`Hit from cloudfront` on every request, `Age` 41 → 45 s, ≈20 ms, POP BNE50
`/api/leaderboard?limit=10` (10 s TTL)	ALB → Lambda → DynamoDB	First request **Miss, 615 ms**; then **Hit, 18–31 ms** with `Age` 1 → 4 s
`index.html` (no-cache)	S3	`RefreshHit`: revalidated each time, so new deployments go live immediately

Content hashing makes the long TTL safe: a new build produces new file names, and only `index.html` is invalidated on deploy. The cached API means the leaderboard costs at most one Lambda invocation and one DynamoDB query per 10 s per edge location, however many players open the join screen.

4.8 Testing

- **Unit (deterministic, seeded PRNG):** movement constants, turn rate, path spacing, growth, eating, collision and boundary death, food drop, NPC fill for 1–4 humans, lobby timing, ROOM_FULL, late join, world completion, respawn, area-of-interest culling, delta encoding, drain, signed IDs, message validation, rate limiting, publisher retry/bounds/flush. Lambda: idempotency, poison messages, partial batch failure, transient errors, commutative world events, DynamoDB command shapes (no Scan), API validation.
- **Integration:** real `ws` clients against the real server, covering shared world, 50 entities, forged and malformed input, ping, ROOM_FULL close code, join timeout and graceful shutdown.
- **Smoke:** `tests/smoke.sh` checks the deployed system through CloudFront, including a real WebSocket session: 9/9 checks passed after the final deployment **[E2]**.
- **Load / scaling / resilience:** §4.5–4.6.

Result: **50/50 automated tests pass** (37 server, 13 Lambda) **[E18]**.

5 Cost and sustainability

5.1 Measured cost drivers

Unit prices are on-demand Sydney / CloudFront-Australia rates read from the AWS public price-list files on 19 Oct 2026. (The Price List API is denied to the student role; free tier is ignored.) Resource quantities come from the deployment and the load tests.

Driver	Measured	Unit price	Cost per player-hour
Fargate task (0.25 vCPU, 0.5 GB, ARM)	1 task per ≤4 humans (×1/0.75 headroom)	$0.03885 vCPU-h + $0.00426 GB-h = **$0.011843/task-h**	≈ $0.0039
Snapshot egress via CloudFront	≈105 KiB/s per player before optimisation; **21.2 KiB/s deployed** (0.078 GB/h)	$0.114/GB (AU, first 10 TB)	≈ $0.042 → **≈ $0.009**
ALB processed bytes (1 LCU = 1 GB/h)	same bytes as egress	$0.008/LCU-h (+ $0.0252/h fixed)	≈ $0.0004 after compression
Events (SQS + Lambda + DynamoDB)	a few events per real player-minute, batched by 10	$0.40/M SQS, $0.20/M Lambda, $0.71/M WRU	< $0.0002

Two conclusions shaped the implementation. First, **egress, not compute, dominates**. That justified delta encoding, half-resolution bodies and permessage-deflate (§4.3), which together cut wire bytes by ~80 % and took the variable cost per player-hour from ≈ $0.047 to ≈ $0.013. Second, the per-player compute cost is fixed by the world size (one 0.25 vCPU task per four seats), so the occupancy target is the main cost lever: 75 % keeps one empty world in four as joining headroom.

5.2 Scenario estimate

For a modest public deployment averaging **60 concurrent humans** (peak 200), 24×7:

```
average worlds needed  = ceil(60 / 4) = 15        running tasks at 75 % target ≈ ceil(60 / 3) = 20
peak (200 humans)      = ceil(200 / 4) = 50 worlds → ≈ 67 tasks (MaxTasks raised from the demo cap of 6)
```

Component	Basis	USD / month
Fargate	20 tasks × $0.011843 × 730 h	172.9
ALB	$18.4 fixed + 4.7 LCU × $0.008 × 730	45.8
CloudFront egress	60 × 0.078 GB/h × 730 h = 3,416 GB × $0.114 (deployed encoding)	389.4
SQS, Lambda, DynamoDB	~1.3 M requests, ~0.5 M invocations, ~6.5 M WRU	≈ 7
CloudWatch	11 custom metrics, Container Insights, ~9 alarms, ~10 GB logs	≈ 14
ECR, Secrets Manager, S3		≈ 0.5
Total	**≈ $0.014 per player-hour**	**≈ 630**

Before the bandwidth optimisations the same scenario would cost ≈ $2,090/month (egress ≈ $1,750 at ≈0.37 GB per player-hour). The minimum floor, at night with one idle world, is ≈ $40/month (ALB + one task + monitoring). The exact AWS Pricing Calculator inputs are in Appendix D. ⟦PRICING_CALCULATOR_LINK: create the public estimate from Appendix D and paste the link here⟧

Not included: the account's pre-existing NAT gateways, used only for image pulls and AWS API calls (~100 MB per task start); Container Insights at large fleet sizes; data transfer of the ~40 KB client bundle (cached in browsers for a year).

5.3 Sustainability

- **Capacity follows demand.** Target tracking on occupancy adds worlds only when players need seats and removes idle ones afterwards (scale-in evidence §4.5). The occupancy target trades a small amount of idle capacity (25 %) for players not waiting.
- **Idle worlds do no work.** When the last human leaves, the world object is discarded and the tick does nothing: an idle task runs at ≈3 % CPU versus ≈25 % active [E10, E15]. NPCs never simulate without players.
- **Right-sizing from measurement.** The smallest Fargate size (0.25 vCPU, 0.5 GB) was kept because a full world used ≈20–30 % CPU and ≈25 MB of memory, with p99 tick time well inside the 33 ms budget.
- **Graviton (ARM64)** for both Fargate and Lambda: AWS states better energy efficiency per unit of work than comparable x86, at 20 % lower price here.
- **Less data moved.** Area-of-interest culling, delta encoding, half-resolution bodies and compression cut bytes per player by ~80 % relative to the first version. Moving fewer bytes means less network energy as well as lower cost.
- **Edge caching.** Fingerprinted assets are cached for a year in browsers and at the edge, and the leaderboard for 10 s, so repeat requests never reach S3, the ALB, Lambda or DynamoDB [E6].
- **Managed and serverless** components (Fargate, Lambda, SQS, DynamoDB on-demand) consume nothing while idle, instead of always-on instances sized for peak.

6 Reflection

⟦STUDENT TO PERSONALISE⟧ This section records what actually happened in the project, taken from the repository history, logs and evidence. Because most implementation work was AI-assisted (see the declaration), rewrite it in your own voice and add your own learning before submission. Do not keep claims you can't stand behind.

6.1 Topics that went beyond the practical material

Topic	Initially unknown	What was investigated / what happened	Effect on the design
Scaling signal for stateful worlds	Whether ECS CPU target tracking (the usual starting point) would react to game load.	Measured a full world (4 players + 46 NPCs) at ≈20–30 % of a 0.25 vCPU Graviton task. Per-world CPU is bounded by design (≤4 humans), so a 40–50 % CPU target would never fire.	A custom `WorldOccupancyPct` metric via EMF became the primary target-tracking signal, with CPU kept as a safety net. Scale-out was then reproducible (1→5 tasks).
ECS task scale-in protection	How to stop scale-in from terminating a world with players.	Used the ECS agent endpoint (`$ECS_AGENT_URI/task-protection/v1/state`), enabled when a world starts and renewed every 5 min.	Verified: active tasks protected, idle task not (E12).
ALB draining vs. SIGTERM for WebSockets	Assumed the server's SIGTERM handler would tell clients to reconnect.	The task-stop test showed ECS deregisters the target first, and the ALB closes long-lived WebSockets when the 15 s deregistration delay ends, about 10 s *before* SIGTERM. The `bye` message is never delivered for ALB-terminated connections.	Resilience comes from the client's reconnect path (verified: rejoin within one 5 s sample). Improvement: poll target health from the task to start draining earlier.
Account guardrails and CloudFormation	Which resource types the student role may create and delete.	Hit explicit denies one by one: stacks stuck in `ROLLBACK_FAILED` (undeletable buckets and log groups), early-validation failures, and CloudFormation's Lambda handler failing on `GetFunction` for a not-yet-existing function.	Split "what may not be deleted" out of CloudFormation into idempotent scripts. Imported a retained DynamoDB table into the new stack. Documented every workaround (§3.2).
Bandwidth as a cost driver	Expected compute to dominate.	Public price list + measured bytes showed CloudFront egress (≈$0.042/player-hour before optimisation) far above Fargate (≈$0.004/player-hour).	Delta encoding, half-resolution bodies and permessage-deflate (≈ −80 % measured in AWS). Compression also exposed a shutdown-flush bug, now fixed and tested.
At-least-once, unordered delivery	How to make score aggregation safe under redelivery.	DynamoDB transactions with a conditional idempotency marker; an idempotent "max" for best score; commutative world updates.	Out-of-order delivery happened in production and the data stayed correct (E8).

6.2 What failed and what changed

- The first NPC brain let NPCs die every ~15 s. Death-cause instrumentation (rather than guessing) found head-to-head collisions while chasing the same food, and a targeted fix raised median lifetime to 23 s.
- A growth bug left the newest tail section `undefined` for one tick. It serialised as `null` and was drawn at (0, 0). It was found by checking rendered snake extents in the browser, fixed in `grow()`, and covered by a regression test.
- The first load test gave up on `ROOM_FULL` after ~2 minutes, before auto-scaling could add capacity (~3 min of alarm datapoints + ~1 min of task start). The retry policy became "five fast jittered attempts, then bounded exponential backoff for ~3 minutes", in both the browser and the load tester.
- Two players in the same browser share `localStorage`, hence the same player ID. An early "replace the older connection" rule would have made two tabs evict each other in a loop, so it was removed.

6.3 What I would do differently

- Check account permissions *before* writing templates (a scripted probe of create/delete/tag actions), rather than discovering denies through failed stacks.
- Add predictive or scheduled scaling for known peaks. Target tracking needs ~3 minutes of datapoints, so a sudden rush waits a few minutes for new worlds.
- Use a binary snapshot format and a 10 Hz snapshot rate before reaching for compression, and measure the CPU/egress trade-off on Fargate itself.
- Request a certificate (or use a VPC origin) so the CloudFront→ALB hop is encrypted, and use dedicated security groups once permitted.

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Appendices

Appendix A. Assessment pattern matrix

Evidence IDs (E#) refer to `docs/evidence-register.md`. Report sections: §2 Vendor-agnostic architecture, §3 Vendor-specific implementation, §4 Application implementation, §5 Cost and sustainability.

#	Pattern	Why this application needs it	Implementation	AWS service	How it is tested	Evidence	Report §	Demo
P1	Structured persistence	Scores, deaths and the leaderboard must survive task loss and be queried by key or rank.	Single table + sparse GSI leaderboard. Idempotent transactional writes.	DynamoDB	Unit tests of the worker against an in-memory store. Deployed: die in game, then query the item and the GSI.	E9	§3, §4	1:55–2:35
P2	Object storage	Frontend build and sprites need durable, cheap hosting with no servers.	Private bucket, OAC-only policy, cache headers set per object.	S3	`aws s3 ls`. Direct S3 URL returns 403. CloudFront URL returns 200.	E2	§3	0:00
P3	Edge caching	Static assets and leaderboard reads should not hit the origin every time.	Fingerprinted assets (1-year immutable). `/api/leaderboard` 10 s TTL.	CloudFront	`tests/cache-test.sh` shows `X-Cache: Miss` then `Hit`, and `Age` increasing. CloudFront cache statistics.	E6	§3, §5	1:20–1:55
P4	Realtime communication	15 Hz state to many clients plus low-latency input.	`ws` server, JSON protocol, interest management, client interpolation, ping/RTT.	ALB + CloudFront WebSocket support	Integration test (two clients share a world). Load test RTT percentiles.	E1, E3, E14	§4	0:30–1:20
P5	Async service communication	DB writes must not block the tick loop. Bursts and outages must not lose results.	Batched `SendMessageBatch` with bounded retry. Graceful flush on SIGTERM.	SQS (+ DLQ)	Unit test of publisher retry. Deployed: queue metrics plus worker logs.	E7, E8	§3, §4	1:55–2:35
P6	Container compute	Long-running stateful simulation holding WebSockets.	Node 22 Alpine image, non-root, healthcheck, arm64.	ECS Fargate, ECR	`docker run` local test. ECS running tasks.	E10, E12	§3	3:50
P7	Serverless compute	Event consumption and the read API are bursty and should scale to zero.	Event worker (SQS ESM, partial batch failure). Leaderboard API (ALB target).	Lambda	Unit tests. CloudWatch invocations/errors. Lambda logs.	E8, E15	§3	2:00
P8	Non-Lambda auto-scaling	More humans need more worlds (`ceil(humans/4)`).	Target tracking on `WorldOccupancyPct` (custom EMF metric, 75 %) and CPU (50 %). ECS task scale-in protection for active worlds.	ECS Service Auto Scaling	14-player load test: desired 1→2→3→4→5 in 7 min. Scale-in after load.	E10, E11, E11b, E12, E15	§3, §4.5	2:35–3:50
P9	Load distribution (dispatcher push)	New connections must spread across worlds. Full worlds must be skipped.	ALB round-robin + `ROOM_FULL` jittered retry.	ALB	Load test reports the distribution of clients per `taskId`. Healthy targets.	E13, E14	§3, §4	3:50–4:30
P10	Stateful authoritative world	Fairness and integrity: clients only send intent.	30 Hz deterministic tick, collisions on a spatial grid, server-side scoring.	ECS task memory	Unit tests (movement, growth, death, food). Integration test that forged messages are ignored.	E1, E4	§2, §4	0:30
P11	Horizontal scaling by independent worlds	Avoid distributed realtime state.	One world per task. No cross-task sync.	ECS	Scale-out creates additional worlds with distinct `worldId` / `taskId`.	E11, E12, E14	§2	3:00
P12	Event-driven processing	Durable reaction to game outcomes.	`PLAYER_DIED`, `SESSION_ENDED`, `WORLD_STARTED`, `WORLD_COMPLETED` events with `eventId`.	SQS, Lambda	Logs show eventId flow from server to worker to DynamoDB.	E7–E9	§4	2:10
P13	Monitoring and observability	Scaling decisions and failures must be visible.	JSON logs, EMF custom metrics, dashboard, alarms.	CloudWatch	Dashboard screenshots. `aws cloudwatch get-metric-data` exports.	E15	§3	throughout
P14	Retries and error handling	Transient AWS errors, duplicates, poison messages.	Bounded exponential backoff in the publisher. SQS redrive (3 attempts) to DLQ. Idempotency markers.	SQS, Lambda, DynamoDB	Inject a malformed event and watch it land in the DLQ. Duplicate eventId applied once (unit test).	E7, E16	§4	optional
P15	Microservice separation	Simulation, persistence and read API scale and fail independently.	Game service (ECS), event worker (Lambda), leaderboard API (Lambda).	ECS, Lambda	Each deploys separately. Failure of one does not stop gameplay.	E8, E13	§2	—
P16	Public vs private access	Minimise attack surface.	Only CloudFront is public. S3 is OAC-only. ALB forwards only requests with CloudFront's secret header (default 403). Tasks in private subnets with no public IP. (Dedicated SGs not permitted in the account.)	CloudFront, S3, ALB, VPC subnets	Direct S3 → 403. Direct ALB without/with wrong header → 403. Tasks have private IPs only.	E2, E19	§3.2–3.3	—
P17	Managed services	Single developer, no ops team.	Fargate, Lambda, DynamoDB, SQS, CloudFront are all managed.	many	—	—	§2, §5	—

P18	Resilience / reconnect	Tasks can die or be replaced.	Client backoff + jitter reconnect. Server graceful SIGTERM drain. ECS + ALB health replacement.	ECS, ALB	Stop a task during play: client reconnects to another world. ECS starts a replacement.	E16	§4	4:00
P19	Configuration management	Same image across environments. Secrets outside the image.	Environment variables in the task definition. SSM SecureString for the token secret. CloudFormation parameters.	SSM, ECS, CloudFormation	Inspect the task definition. Image contains no secrets (`docker history`).	E10	§3	—
P20	Infrastructure as code	Reproducibility.	CloudFormation stacks + deploy scripts.	CloudFormation	Stack create/update from a clean state.	inventory	§3	—

Appendix B. Evidence register

Every report claim maps to an item here. Files are under `evidence/`. Times are UTC (Brisbane = UTC+10).

"Screenshot" items are listed in `evidence/screenshots-needed.md` for the student to capture.

Status: ✅ captured · ⏳ pending capture · 📸 needs a console screenshot.

ID	What it proves	Command / test	When (UTC)	AWS resource	Result	Screenshot?	Report §	Video
E1	Local multiplayer: two real browser clients share one authoritative world	`npm run dev`, two browser tabs, `GET /status`. Integration test `server/test/integration.test.js`	2026-10-19 01:01	local	✅ `/status`: `humans 2, npcs 48, worldId Maclan-1`. Both tabs on `world Maclan-1`. Integration test: both clients get the same `worldId`, 50 entities	Optional	§4	—
E2	Deployed frontend over CloudFront; S3 is private	`tests/smoke.sh`, `curl` direct S3	2026-10-19 02:01	CloudFront `d3riy98bj197ur.cloudfront.net`, S3 `n5528712-a3-frontend-901444280953`	✅ index 200 via CloudFront. Direct S3 URL **403**	📸 app in browser	§3	0:00
E3	WebSocket multiplayer through CloudFront → ALB → ECS	`load-tests/websocket-load.js --clients 2` via `wss://…/ws`	2026-10-19 02:01	ALB `n5528712-a3-alb`, service `n5528712-a3-game`	✅ 2/2 joined, RTT p50 19 ms / p95 61 ms, snapshot interval p50 67 ms	📸 two windows	§4	0:30
E4	50-entity world (humans + NPC fill)	unit tests `room.test.js` ("NPC fill is 50 – humans for 1..4 humans"); `/status`	2026-10-19	ECS task	✅ 1→49, 2→48, 3→47, 4→46 NPCs (tests). Deployed `/status` entities 50	—	§4	0:30
E5	Lobby ≤10 s; starts immediately at 4 humans; ROOM_FULL for the 5th	unit tests + integration test; deployed lobby	2026-10-19	ECS task	✅ starts by 10 s (timer) or instantly when full. 5th client: `ROOM_FULL`, close 4001	📸 lobby countdown	§4	0:15
E6	CloudFront cache MISS → HIT	`tests/cache-test.sh … 5` → `evidence/aws/E6-cache-test.txt`	2026-10-19 02:02	CloudFront, POP BNE50	✅ asset Hit (Age 41→45 s, ~20 ms). API Miss 615 ms → Hit 18–31 ms. index.html RefreshHit (no-cache)	📸 optional	§3, §5	1:20
E7	SQS carries game events; queue drains; DLQ catches poison messages	queue depths; `E16-dlq-test.txt`; `E7-queue-depths-*.txt`	2026-10-19 02:02–02:23	SQS `n5528712-a3-events`, `…-dlq`	✅ events sent and deleted, depth 0 after runs. Poison message → DLQ (see E16)	📸 S8	§3, §4	2:00
E8	Lambda consumes events (idempotent, partial batch failures)	CloudWatch Logs `/aws/lambda/n5528712-a3-event-worker`	2026-10-19 02:01	Lambda `n5528712-a3-event-worker`	✅ `event_processed` for PLAYER_DIED, WORLD_STARTED, WORLD_COMPLETED, SESSION_ENDED (`result: applied`)	📸 log lines	§4	2:10
E9	DynamoDB stores results; leaderboard via GSI Query	`aws dynamodb query --index-name GSI1`; `/api/leaderboard`	2026-10-19 02:02	DynamoDB `n5528712-a3-game`	✅ ranked items from GSI1 (no Scan). API serves the same	📸 table items	§4	2:20
E10	Single-task baseline	`tests/watch-scaling.sh 1 1` → `E10-baseline-single-task.txt`	2026-10-19 02:04	ECS service	✅ desired 1, running 1, occupancy 0 %, CPU 2.9 %, min 1 / max 6	—	§3	2:35

E11	ECS Service Auto Scaling scale-out under load	`load-tests/websocket-load.js --clients 14 --duration 1500` + `tests/watch-scaling.sh` → `E11-scaling-timeline.txt`, `E11-scaling-activities-*.txt`, charts	2026-10-19 02:07–02:32	ECS service + Application Auto Scaling (occupancy policy)	✅ desired 1→2 (12:10:15), →3 (12:12:15), →4 (12:15:15), →5 (12:17:15 AEST). Occupancy 100 % → 70 % steady. New tasks healthy 30–45 s after each decision. Scale-in: see E11b	📸 S3, S4, S6	§3, §4.5	2:35–3:50
E12	Multiple running tasks = independent worlds; scale-in protection on active worlds	`E12-running-tasks.txt`, `E12-task-scale-in-protection.txt`, `E11b-protection-after-load.txt`	2026-10-19 02:17–02:33	ECS tasks	✅ 5 RUNNING tasks in 3 AZs. 4 active worlds protected, idle task unprotected. All released after load	📸 S5	§4.5	3:30
E13	ALB healthy targets across AZs	`E13-target-health-5-tasks.txt`	2026-10-19 02:17	target group `n5528712-a3-game-tg`	✅ 5 targets (2a ×2, 2b, 2c ×2), healthy after registration	📸 S7	§3	3:50
E14	Load-test results (connections, RTT, distribution)	`E14-load-test-scaleout.json/.log`; local `evidence/local/E14*`	2026-10-19 02:07–02:32	CloudFront → ALB → ECS	✅ 931 attempts, 0 errors, 913 ROOM_FULL retries, 0 gave up, 14/14 seated across 5 tasks (4/4/4/2 + headroom). RTT p50 16 / p95 29 / p99 68 ms (8,769 samples). Snapshot interval p50 67 / p95 71 ms. 262,605 snapshots	—	§4.5	3:00
E15	CloudWatch metrics (CPU, memory, occupancy, tasks, tick, SQS, Lambda, DynamoDB)	`collect-evidence.sh` → `E15-cloudwatch-metrics-*.csv/json`; `report/figures/chart-*.svg`	2026-10-19	CloudWatch	✅ active task CPU 20–30 %, idle ≈3 %, memory ≈10 %. Tick p50 1.3 ms, p99 avg 2.3 ms (max 13.4)	📸 S3, S13	§4.5, §5	throughout
E16	Resilience: DLQ after 3 attempts; reconnect after task loss; ECS replacement; event flush	`E16-dlq-test.txt`, `E16-task-failure-test.txt`	2026-10-19 02:08–02:23	SQS DLQ, ECS	✅ poison message received 3× (60 s apart) → DLQ → alarm. Task with 4 players stopped: players rejoined other worlds within one 5 s sample. Replacement healthy 47 s after deregistration. 206 events flushed, 0 dropped	📸 S9	§4.6	4:00
E17	Cost measurements (task size, utilisation, bandwidth per player)	`E17-server-snapshot-bytes-uncompressed.txt` (before), `E17-deployed-bandwidth-4players.json` (after), local `E17-deflate-*.json`; `docs/cost-estimate.md`	2026-10-19 02:20 / 07:54–08:04	CloudWatch, CloudFront	✅ before ≈105 KiB/s/player uncompressed; deployed 21.2 KiB/s/player on the wire (−80 %), RTT p50 17 ms. ≈ $0.014 per player-hour	—	§5	—
E18	Automated tests pass	`npm test` → `evidence/local/automated-tests.txt`	2026-10-19	—	✅ 37 server + 13 Lambda tests pass	—	§4	—
E11b	Scale-in after load (protection released, idle worlds removed)	`E11b-scale-in-timeline.txt`, `E11b-scaling-activities.txt`, `E11b-protection-after-load.txt`	2026-10-19 02:32–02:59	ECS + Application Auto Scaling	✅ protection released at 12:32. Desired 5→4 (12:47:07), →3 (12:51:20), →2 (12:55:20), →1 (12:59:20 AEST)	📸 S3	§4.5	—
E19	Security controls	`E19-security-checks.txt`; protocol tests	2026-10-19 02:22	ALB, S3, ECS	✅ ALB without/with wrong header → 403 (HTTP and WS upgrade). Direct S3 → 403. HTTP → 301 HTTPS. Public access block on. All tasks private IP only	—	§3.3	—

Appendix C. Acceptance test results

Run on 2026-10-19 against the deployed system (`https://d3riy98bj197ur.cloudfront.net/`) and the repository at

the commit noted in each evidence file. PASS = requirement met with evidence; PARTIAL = met with a stated limitation; FAIL = not met.

Functional requirements

ID	Requirement	Result	Evidence
F1	Anonymous display name (sanitised, ≤16 chars) and join	**PASS**	protocol tests; deployed join (E3)
F2	Steer with mouse/arrow keys, boost with SPACE/click	**PASS**	client; local browser test
F3	Eat food and grow (+1 section, ×1.01 scale)	**PASS**	`snake.test.js`, `world.test.js`
F4	Death on head-edge contact with another body or the arena edge; body becomes food	**PASS**	`world.test.js` (collision, boundary, food drop)
F5	Dead human sees score and respawns (3 s or on request)	**PASS**	`room.test.js` respawn test; death screen in browser
F6	50 snake slots, ≤4 humans, NPCs = 50 − humans	**PASS**	`room.test.js` (1→49 … 4→46); deployed `/status` entities 50
F7	Lobby ≤10 s; starts immediately with 4 humans	**PASS**	`room.test.js`, integration test; WORLD_STARTED `lobbyWaitMs`
F8	Late join replaces an NPC; 5th human gets ROOM_FULL and is routed elsewhere	**PASS**	unit + integration tests; load test: ROOM_FULL clients joined new tasks (E11/E14)
F9	Score, in-world top 10 and global top 10 visible	**PASS**	browser UI; `/api/leaderboard` (E9)
F10	Connection status; bounded jittered reconnect	**PASS**	task-failure test: 4 players rejoined other worlds within one 5 s sample (E16)
F11	Durable results feed a global leaderboard	**PASS**	SQS → Lambda → DynamoDB logs and items (E7–E9)
F12	Multiple real browsers share a world	**PASS**	local two-browser test (E1); deployed screenshots pending (S1)

Cloud / non-functional requirements

Requirement	Result	Evidence
CloudFront serves the frontend from private S3	**PASS**	E2 (direct S3 403)
Cache hit demonstrated	**PASS**	E6: asset Hit, Age 41→45; API Miss 615 ms → Hit ~20 ms; BNE50
ECS/Fargate hosts the game server	**PASS**	E10, E12
ALB distributes connections	**PASS**	E14: 14 players distributed 4/4/4/2 across tasks; E13 healthy targets in 3 AZs
ECS Service Auto Scaling works, with real scale-out evidence	**PASS**	E11: desired 1→2→3→4→5 (12:10:15 → 12:17:15 AEST) under synthetic gameplay load
Scale-in after load	**PASS**	E11b: 5→4→3→2→1 between 12:47 and 12:59 AEST, after 15 min below target
Scale-in protection for active worlds	**PASS**	E12-task-scale-in-protection.txt: 4 active tasks protected, idle task not
SQS async processing, Lambda consumer, DynamoDB storage	**PASS**	E7, E8, E9
Retries / DLQ	**PASS**	E16-dlq-test.txt: 3 receives 60 s apart → DLQ → alarm
Idempotent consumer	**PASS**	`lambda/test/worker.test.mjs`, `dynamoStore.test.mjs`
Reconnect after server loss; ECS replaces the task	**PASS**	E16-task-failure-test.txt
Graceful shutdown flushes events	**PASS**	E16: 206 published, 0 dropped on SIGTERM
CloudWatch evidence (metrics, logs, dashboard, alarms)	**PASS**	E15 exports; dashboard `n5528712-a3`
Automated tests	**PASS**	37 server + 13 Lambda tests (E18)
Load test tool works	**PASS**	E14
Infrastructure reproducible	**PARTIAL**	4 CloudFormation stacks + idempotent scripts. Buckets, log groups and Lambdas are created by scripts because of account deny policies (documented)
Required tags on taggable resources	**PASS**	`describe-stack.sh` inventory: all A3 stacks, ECS, ALB, Lambda, SQS, DynamoDB, ECR, S3, logs, alarms, secret, parameter, scalable target + CloudFront distribution carry `purpose=assessment 3` and `qut-username`. Non-taggable: OAC, cache policy, dashboard, bucket policy
Security: validation, rate limiting, no client trust	**PASS**	protocol/integration tests
Security: private origin	**PARTIAL**	ALB secret header + private tasks verified (E19). The SG can't be restricted and the origin hop is HTTP (no SG/ACM permissions)
Least-privilege IAM	**PARTIAL**	pre-provisioned roles only (iam:CreateRole denied); policies not inspectable
Log retention ≤7 days	**FAIL (account)**	`logs:PutRetentionPolicy` denied to the student role; retention left at default

Definition of done (from the project brief)

Item	Status	Note
original gameplay assets reused appropriately	✅	all 7 PNGs, Phaser CE 2.6.2, original constants/rules (docs/current-state.md §4)
authoritative server works	✅	37 server tests; deployed

two real clients can play together	✅	E1 (local, two browsers), E3/E14 (deployed). Deployed two-window screenshot: S1
maximum normal humans/world = 4	✅	ROOM_FULL tests; load test 4/4/4/2
world reaches 50 total entities using NPC fill	✅	tests + `/status`
lobby wait is never longer than 10 seconds	✅	tests; `lobbyWaitMs` ≤ 10,033
CloudFront serves frontend	✅	E2
cache HIT can be demonstrated	✅	E6
ECS/Fargate hosts game server	✅	E10/E12
ALB distributes connections	✅	E13/E14
ECS Service Auto Scaling works	✅	E11
real scale-out evidence exists	✅	E11 timeline + scaling activities + charts
SQS async processing works	✅	E7
Lambda consumes events	✅	E8
DynamoDB stores results	✅	E9
leaderboard works	✅	`/api/leaderboard` + UI
reconnect behaviour works	✅	E16
CloudWatch evidence exists	✅	E15, dashboard
automated tests pass	✅	50/50
load test works	✅	E14
infrastructure is reproducible	⚠️ PARTIAL	CloudFormation + scripts; some resources by script due to account denies
required AWS resources are correctly tagged	✅	inventory
submission YAML contains real deployed resources	✅	`submission/submission-infra.yml` (generated from live ARNs, YAML-validated)
vendor-neutral architecture diagram exists	✅	`report/figures/vendor-neutral.svg`
AWS architecture diagram exists	✅	`report/figures/aws-architecture.svg`
AWS Pricing Calculator inputs are prepared	✅	`docs/cost-estimate.md` (public link must be created by the student)
evidence register is complete	✅	`docs/evidence-register.md` (console screenshots listed in `evidence/screenshots-needed.md`)
demo plan is complete	✅	`docs/demo-video-plan.md`
report describes the actual deployed system	✅	`report/`
report does not invent evidence	✅	every figure traced to `evidence/`
AI declaration is truthful and awaiting user review	⏳	marked ⟦STUDENT TO REVIEW⟧ on the title page

Appendix D. AWS Pricing Calculator inputs

AWS Pricing Calculator: exact inputs

Create an estimate at https://calculator.aws/ (Region **Asia Pacific (Sydney)**), add these services, then

Save and share and paste the public link into the report. *(The link must be created by you; it is not invented here.)*

- **AWS Fargate**: OS Linux, CPU architecture **ARM**, number of tasks **20** per day, average duration **24 hours**

(or "730 hours per month"), vCPU **0.25**, memory **0.5 GB**, ephemeral storage 20 GB (default, no charge).

- **Elastic Load Balancing**: Application Load Balancer, 1 ALB, processed bytes **4.7 GB per hour** (EC2/IP targets),

average new connections **1 per second**, average connection duration **600 s**, rule evaluations default.

- **Amazon CloudFront**: Australia, data transfer out to internet **3,416 GB/month**, number of HTTPS requests

100,000/month, data transfer out to origin **20 GB/month**.

- **AWS Lambda** (two entries or one combined): ARM, **200,000** requests/month, **50 ms**, **256 MB**;

ARM, **300,000** requests/month, **20 ms**, **128 MB**.

- **Amazon SQS**: Standard queue, **1.3 million** requests/month.
- **Amazon DynamoDB**: On-demand, Standard table class, storage **1 GB**, writes **6.5 million**, reads **0.5 million**.
- **Amazon CloudWatch**: **25** custom/other metrics, **9** standard alarms, **10 GB** logs ingested (standard), **10 GB** stored, **1** dashboard (first 3 free).
- **Amazon ECR**: **0.3 GB** storage. **AWS Secrets Manager**: **1** secret, 10,000 API calls.

Appendix E. Repository and reproduction

Source: the `a3-cloud-multiplayer` branch of https://github.com/CY3639/slither.io-clone. Key paths: `server/` (game server, Dockerfile), `client/` (Phaser client and build), `lambda/` (event worker and leaderboard API), `infra/` (CloudFormation and scripts), `load-tests/`, `tests/`, `evidence/`.

Deploy with `AWS_PROFILE=default infra/scripts/deploy.sh`; run locally with `npm run install:all && npm run dev`; test with `npm test`. Teardown steps are in `infra/CLEANUP.md`.